EEG-Based Analysis of Alzheimer's Disease

<u>Related Works Table:</u>

Title	Authors	Year	Dataset	Application Area	Method/Model	Metrics	Strengths	Limitations	Research Gap	Relation to Work	Type of GNN
Graph neural networks for electroencephalogram analysis: Alzheimer's disease and epilepsy use cases	Sergi Abadal et al.	2025	Private AD dataset (63 AD, 63 MCI, 63 HC)	AD diagnosis and seizure discrimination	**Graph Transformer** architecture	Ternary AD Accuracy: **89.02%**; Seizure Accuracy: **96.95%**	Captures non-Euclidean relations; eliminates the need for pre-defined edge weights; high explainability	Higher number of trainable weights compared to vanilla GCN	Prior works were limited to simple GNNs or binary classification; they lacked large-scale design explorations	Directly validates the use of **Graph Transformers (attention-based GNNs)** for AD detection	**Graph Transformer**
Variational Mixture of Graph Neural Experts for Alzheimer's Disease Recognition across Frequency Bands	Jun-En Ding et al.	2025	Open AD (88 subjects) and Session-based AD (123 subjects)	AD/FTD subtyping and dementia staging	Variational Mixture of Graph Neural Experts (**VMoGE**)	AUC for HC vs AD: **0.89**	Models' latent uncertainty; uses expert specialized frequency bands; provides neurophysiological interpretable markers	Single-modality limitation; AD neurodegeneration is multi-level and complex	Prior methods were hindered by full-band frequency analysis and lacked comprehensive evaluations with clinical indicators	Implements a **variational gating mechanism** to integrate multi-band EEG features	**Variational GNN / Mixture-of-Experts**
Adaptive Gated Graph Convolutional Network for Explainable Diagnosis of Alzheimer's Disease Using EEG Data	Dominik Klepl et al.	2023	20 AD patients and 20 Healthy Controls (Sheffield memory clinic)	Explainable AD diagnosis	Adaptive Gated Graph Convolutional Network (**AGGCN**)	Accuracy (Eyes-Closed): **89.1%**; AUC: **0.895**	Adaptively learns graph structures; gating filters info across spatial scales; explainable predictions via ASAP pooling	Small sample size; short window lengths may lose low-frequency signal info	GNN-based AD diagnosis remains relatively unexplored; prior studies used simple GNNs without gating	The **gating mechanism** adaptively weights spatial scales aligning with our requirement	**Gated Graph Convolutional Network**
Flexible and Explainable Graph Analysis for EEG-based Alzheimer's Disease Classification	Jing Wang et al.	2023	108 AD patients and 15 Healthy Control(Taipei Veteran General Hospital)	AD classification across different dementia stages (CDR 0.5–2+)	Flexible **GGCN** with MOTPE hyperparameter tuning	AUC: **>0.90** (Moderate to Severe Dementia); high precision/recall	Highly flexible; optimizes block numbers for clinical metrics; reveals regional connectivity differences in brain lobes	Specificity variability in mild cases	Need for better interpretability of adjacency matrices to reveal regional connectivity differences	Focuses on **flexibility and explainability** using gated mechanisms	**Gated Graph Convolutional Network**
Comorbidity-based framework for AD classification using GNNs	Ferial Abuhantash et al..	2024	ADNI (1331 subjects) & AIBL (858 subjects)	Early AD prediction using EHR data	Neighborhood Patient Graph with **ChebConv/GAT**	Multi-class Acc: **0.98**; AD/CN Acc: **0.99**	Innovative use of EHR comorbidity data; cost-effective alternative to imaging	Sensitive to feature assignment; performance modulations if comorbidities removed	EHR-based comorbidity data for AD GNN classification largely overlooked	Provides performance benchmarks for **GAT (Graph Attention Network)** in AD classification.	**GAT**, ChebConv, GraphSAGE
A novel hybrid model in the diagnosis and classification of Alzheimer's disease: Deep ensemble learning approach	Majid Nour et al.	2024	Combined public datasets from Florida State University (104 AD, 36 HC)	AD and Healthy Control classification using raw EEG	Deep Ensemble Learning (**DEL**) with five 2D-CNN classifiers	Average Accuracy: **97.9%**	Achieves state-of-the-art accuracy; automated feature extraction; robust against overfitting	Lacks interpretability; further effort needed to understand clinical application impacts	Limited previous use of deep ensemble learning in EEG-based AD classification	Uses the **exact same public dataset** we have chosen; provides the **CNN-based baseline** we need	**N/A** (CNN-based Ensemble)